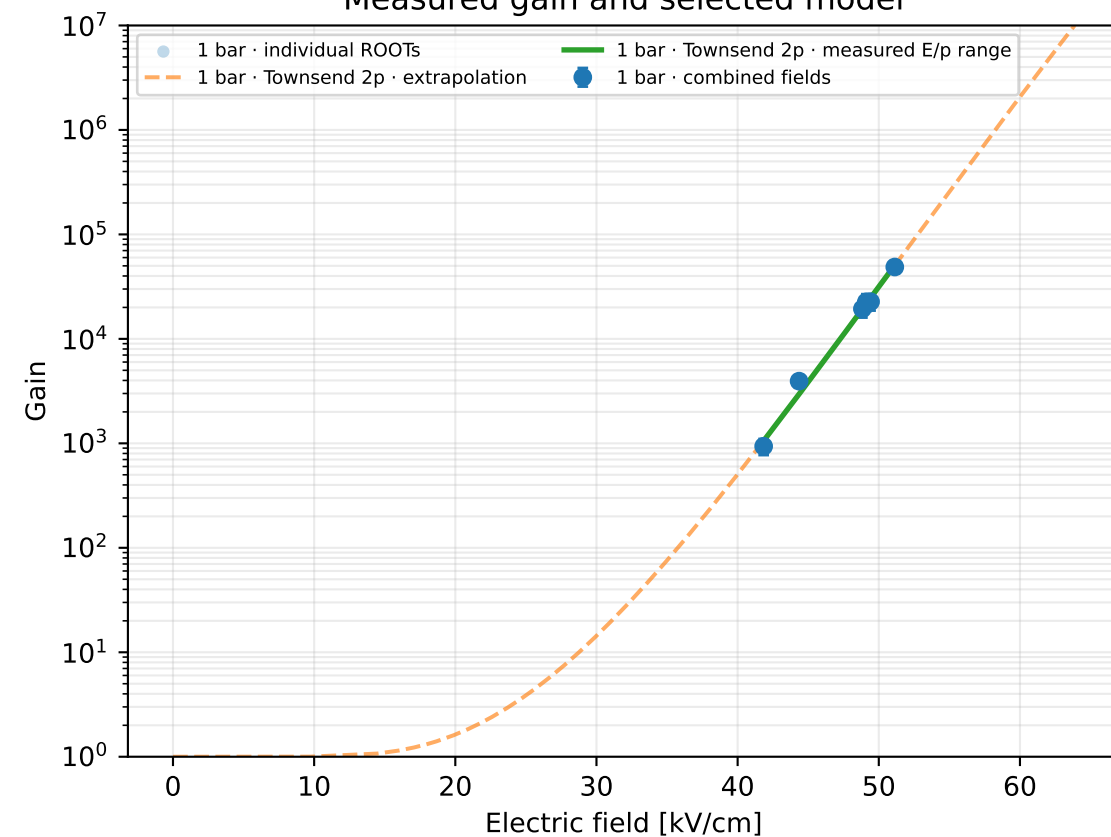
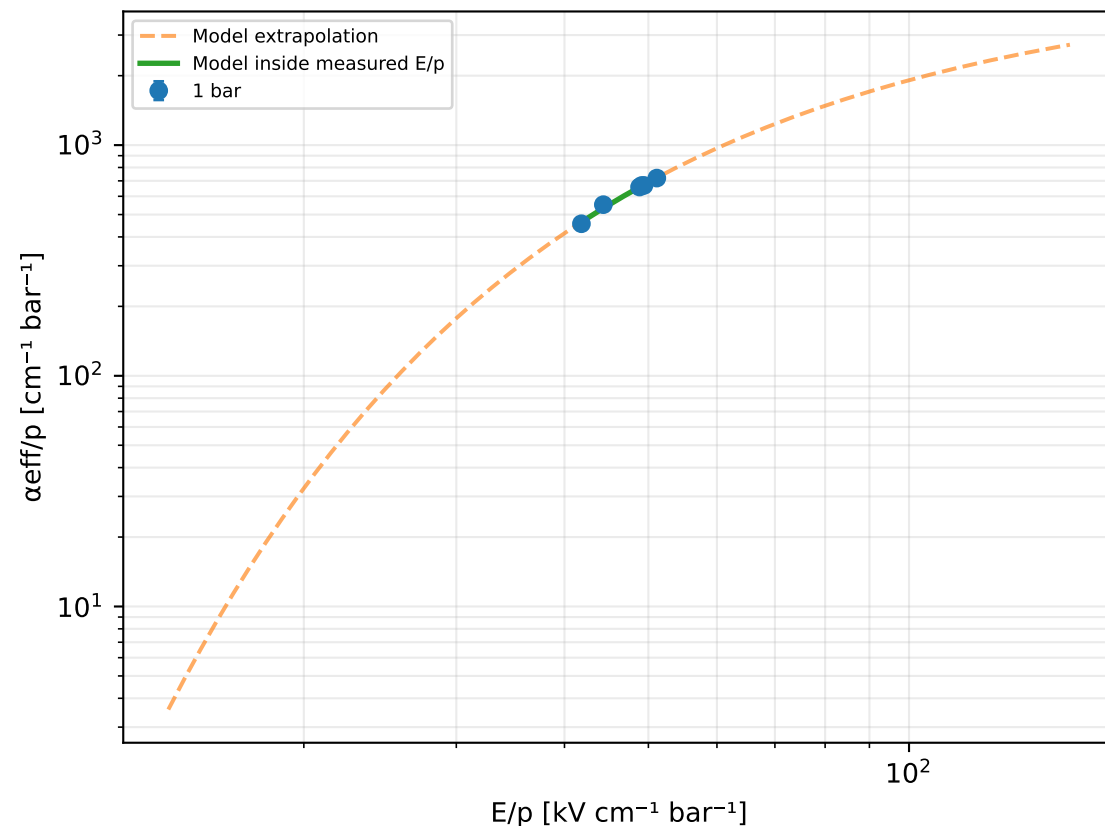


ArIsoCF4 · Ar 87 % / Iso 10 % / CF4 3 % · gap 0.15 mm

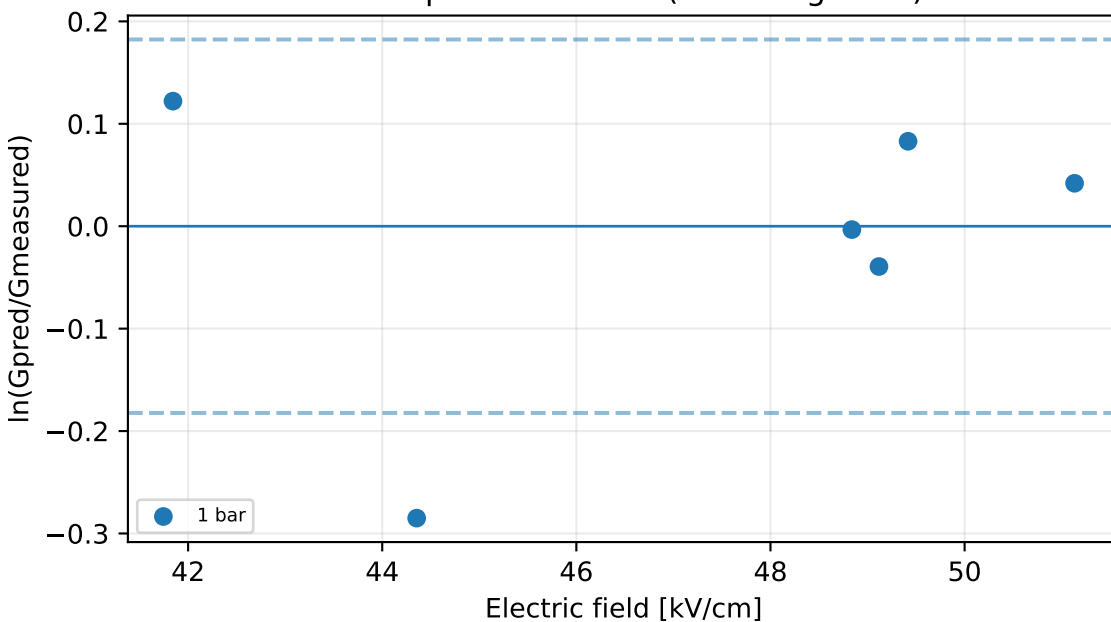
Measured gain and selected model



Reduced Townsend fit

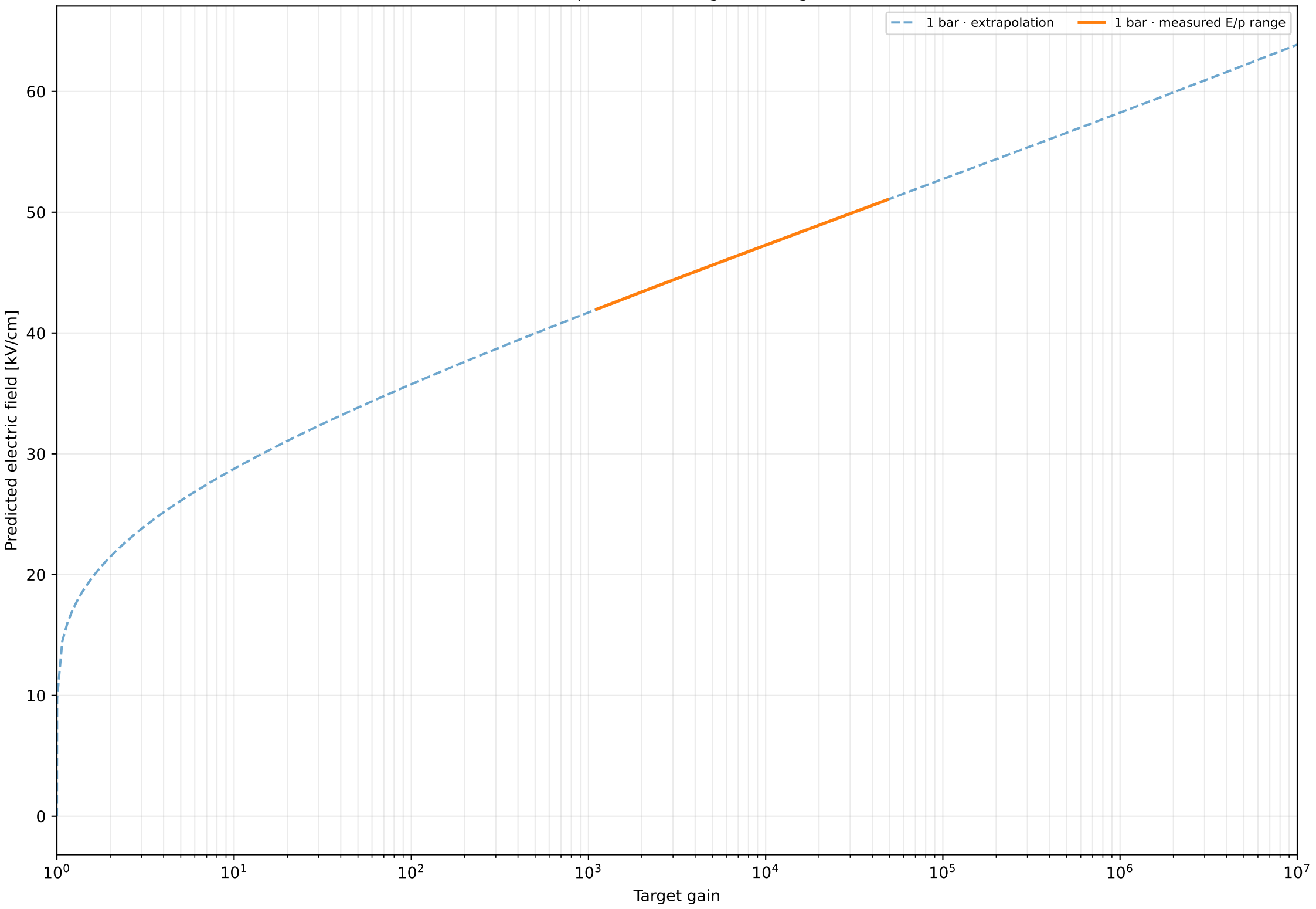


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 6
 Unique physical points: 6
 Pressures: 1 bar
 Selected model: Townsend 2p
 Status: accepted
 Training RMSE: $\times 1.017$
 Cross-validation median: $\times 1.092$
 Cross-validation worst: $\times 1.646$
 Reduced-field range: 41.84–51.13 kV cm⁻¹ bar⁻¹
 A=5290.7, B=101.81, m=0, n=1
 Covariance condition: 1.53e+03

Field extrapolation from gain 1 to gain 10^7



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
1	41.8443	41.8443	937.25	166	456.197	20	1
1	44.3549	44.3549	3938.4	534	551.902	20	1
1	48.8387	48.8387	19379	3.4e+03	658.13	20	1
1	49.1177	49.1177	22597	4.22e+03	668.371	20	1
1	49.4175	49.4175	22689.4	4.05e+03	668.644	20	1
1	51.1329	51.1329	48744.25	6.58e+03	719.623	20	1

Progressive model selection
The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-41.2	×1.09	×1.65	yes	—